十 三重积分与曲线、曲面积分

一、选择题

1. (C). $\bar{z}=\dfrac{\iiint\limits_{\Omega} z\,\mathrm{d}V}{\iiint\limits_{\Omega}\mathrm{d}V}$;

由 $\iiint\limits_{\Omega} z\,\mathrm{d}V=\int_0^2 z\,\mathrm{d}z\iint\limits_{x^2+y^2\leqslant z^2}\mathrm{d}x\,\mathrm{d}y=\pi\int_0^2 z^3\,\mathrm{d}z=4\pi$, $\iiint\limits_{\Omega}\mathrm{d}V=\int_0^2\mathrm{d}z\iint\limits_{x^2+y^2\leqslant z^2}\mathrm{d}x\,\mathrm{d}y=\pi\int_0^2 z^2\,\mathrm{d}z=\dfrac{8}{3}\pi$,

得 $\bar{z}=\dfrac{4\pi}{\frac{8}{3}\pi}=\dfrac{3}{2}$. 选(C).

2. (D). $\Phi=\oiint\limits_S (x^3+2y)\,\mathrm{d}y\,\mathrm{d}z+(y^3+2z)\,\mathrm{d}z\,\mathrm{d}x+(z^3+2x)\,\mathrm{d}x\,\mathrm{d}y$

$$=-3\iiint\limits_{\Omega}(x^2+y^2+z^2)\,\mathrm{d}V=-3\int_0^{2\pi}\mathrm{d}\theta\int_0^{\frac{\pi}{2}}\mathrm{d}\varphi\int_0^{2\cos\varphi} r^4\sin\varphi\,\mathrm{d}r$$

$$=-6\pi\int_0^{\frac{\pi}{2}}\sin\varphi\,\mathrm{d}\varphi\int_0^{2\cos\varphi} r^4\,\mathrm{d}r=-\frac{192\pi}{5}\int_0^{\frac{\pi}{2}}\sin\varphi\cos^5\varphi\,\mathrm{d}\varphi=-\frac{32\pi}{5}.$$

选(D).

3. (D). 因为曲线 L 关于 $y=x$ 对称,由轮换对称性得

$$I=\oint_L\left(x^{\frac{4}{3}}+x^{\frac{2}{3}}y^{\frac{2}{3}}\right)\mathrm{d}s=\oint_L\left(y^{\frac{4}{3}}+y^{\frac{2}{3}}x^{\frac{2}{3}}\right)\mathrm{d}s,$$

从而 $2I=\oint_L\left(x^{\frac{4}{3}}+x^{\frac{2}{3}}y^{\frac{2}{3}}\right)\mathrm{d}s+\oint_L\left(y^{\frac{4}{3}}+y^{\frac{2}{3}}x^{\frac{2}{3}}\right)\mathrm{d}s=\oint_L\left(x^{\frac{2}{3}}+y^{\frac{2}{3}}\right)^2\mathrm{d}s=2^{\frac{4}{3}}\oint_L\mathrm{d}s$.

$I=2^{\frac{1}{3}}\oint_L\mathrm{d}s$,曲线 L 的参数形式为 $L:\begin{cases}x=2\cos^3 t,\\ y=2\sin^3 t\end{cases}(0\leqslant t\leqslant 2\pi)$,故

$$I=2^{\frac{1}{3}}\oint_L\mathrm{d}s=2^{\frac{1}{3}}\cdot 4\int_0^{\frac{\pi}{2}}\sqrt{x_t'^2+y_t'^2}\,\mathrm{d}t=2^{\frac{1}{3}}\cdot 4\int_0^{\frac{\pi}{2}}\sqrt{36\cos^4 t\sin^2 t+36\sin^4 t\cos^2 t}\,\mathrm{d}t$$

$$=2^{\frac{1}{3}}\cdot 24\int_0^{\frac{\pi}{2}}\sin t\,\mathrm{d}(\sin t)=12\sqrt[3]{2}.$$

选(D).

4. (D). $P=\dfrac{y}{(x-1)^2+4y^2}$, $Q=\dfrac{1-x}{(x-1)^2+4y^2}$,则

$$\frac{\partial Q}{\partial x}=\frac{\partial P}{\partial y}=\frac{(x-1)^2-4y^2}{[(x-1)^2+4y^2]^2}((x,y)\neq(1,0)),$$

令 $L_0:(x-1)^2+4y^2=r^2(r>0$, L_0 在 L 内,逆时针方向), L_0^- 与 L 围成的区域为 D_1, L_0 围成的区域为 D_2,由 $\oint_{L+L_0^-}\dfrac{y\,\mathrm{d}x+(1-x)\,\mathrm{d}y}{(x-1)^2+4y^2}=\iint\limits_{D_1}0\,\mathrm{d}x\,\mathrm{d}y=0$,得

$$I=\oint_L\frac{y\,\mathrm{d}x+(1-x)\,\mathrm{d}y}{(x-1)^2+4y^2}=\oint_{L_0}\frac{y\,\mathrm{d}x+(1-x)\,\mathrm{d}y}{(x-1)^2+4y^2}=\frac{1}{r^2}\oint_{L_0}y\,\mathrm{d}x+(1-x)\,\mathrm{d}y$$

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$$=\frac{1}{r^2}\iint_{D_2}(-1-1)\mathrm{d}x\,\mathrm{d}y=\frac{-2}{r^2}\cdot\pi\cdot r\cdot\frac{r}{2}=-\pi.$$

选(D).

5.(A). $|\boldsymbol{F}|=\sqrt{x^2+y^2}$,$\overrightarrow{OM}=\{x,y\}$,与$\overrightarrow{OM}$垂直且与$y$轴正向夹角为钝角的向量为$\{y,-x\}$,$\boldsymbol{F}^0=\frac{1}{\sqrt{x^2+y^2}}\{y,-x\}$,则$\boldsymbol{F}=|\boldsymbol{F}|\cdot\boldsymbol{F}^0=\{y,-x\}$,

$$\begin{aligned}W&=\int_L y\mathrm{d}x-x\mathrm{d}y=\oint_{L+\overline{AO}}y\mathrm{d}x-x\mathrm{d}y+\int_{\overline{OA}}y\mathrm{d}x-x\mathrm{d}y\\&=-\iint_D(-1-1)\mathrm{d}x\,\mathrm{d}y+\int_0^2 0\mathrm{d}x=2\iint_D\mathrm{d}x\,\mathrm{d}y\\&=2\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^{2\cos\theta}r\,\mathrm{d}r=4\int_0^{\frac{\pi}{2}}\cos^2\theta\,\mathrm{d}\theta=\pi.\end{aligned}$$

选(A).

6.(A). 由对称性与奇偶性,得$I=\iint_\Sigma(x^2+xy)\mathrm{d}S=\iint_\Sigma x^2\mathrm{d}S$,其中

$$\Sigma:z=\sqrt{x^2+y^2}\,(D:x^2+y^2\leqslant 2x),$$

由$\frac{\partial z}{\partial x}=\frac{x}{\sqrt{x^2+y^2}}$,$\frac{\partial z}{\partial y}=\frac{y}{\sqrt{x^2+y^2}}$,得

$$\begin{aligned}I&=\iint_D x^2\cdot\sqrt{1+\left(\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2}\,\mathrm{d}x\,\mathrm{d}y=\sqrt{2}\iint_D x^2\mathrm{d}x\,\mathrm{d}y=\sqrt{2}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^2\theta\,\mathrm{d}\theta\int_0^{2\cos\theta}r^3\mathrm{d}r\\&=4\sqrt{2}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^6\theta\,\mathrm{d}\theta=8\sqrt{2}\int_0^{\frac{\pi}{2}}\cos^6\theta\,\mathrm{d}\theta=8\sqrt{2}\times\frac{5}{6}\times\frac{3}{4}\times\frac{1}{2}\times\frac{\pi}{2}=\frac{5\sqrt{2}\pi}{4}.\end{aligned}$$

选(A).

二、填空题

1. 解 令$\Omega_1=\{(x,y,z)\mid x^2+y^2+z^2\leqslant 4,x\geqslant 0,y\geqslant 0,z\geqslant 0\}$,由对称性,得

$$\begin{aligned}I&=\iiint_\Omega(x^2+4y^2+z^2+4xy-2xz-4yz)\mathrm{d}V\\&=\iiint_\Omega(x^2+4y^2+z^2)\mathrm{d}V=4\iiint_{\Omega_1}(x^2+4y^2+z^2)\mathrm{d}V=4I_1,\end{aligned}$$

由$\iiint_{\Omega_1}x^2\mathrm{d}V=\iiint_{\Omega_1}y^2\mathrm{d}V=\iiint_{\Omega_1}z^2\mathrm{d}V$,得

$$\begin{aligned}I&=24\iiint_{\Omega_1}x^2\mathrm{d}V=8\iiint_{\Omega_1}(x^2+y^2+z^2)\mathrm{d}V\\&=8\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^{\frac{\pi}{2}}\sin\varphi\,\mathrm{d}\varphi\int_0^2 r^4\mathrm{d}r=4\pi\cdot\frac{32}{5}=\frac{128\pi}{5}.\end{aligned}$$

2. 解 设质心坐标为$(\overline{x},\overline{y},\overline{z})$,由对称性,得$\overline{x}=0$,$\overline{y}=0$,$\overline{z}=\dfrac{\iiint_\Omega z\mathrm{d}V}{\iiint_\Omega\mathrm{d}V}$.

$$\iiint_\Omega\mathrm{d}V=\int_0^1\mathrm{d}z\iint_{x^2+y^2\leqslant 1-z}\mathrm{d}x\,\mathrm{d}y=\pi\int_0^1(1-z)\mathrm{d}z=\frac{\pi}{2},$$

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$$\iiint_{\Omega} z\,\mathrm{d}V=\int_0^1 z\,\mathrm{d}z\iint_{x^2+y^2\leqslant 1-z}\mathrm{d}x\,\mathrm{d}y=\pi\int_0^1(z-z^2)\mathrm{d}z=\frac{\pi}{6},$$

故质心坐标为$\left(0,0,\frac{1}{3}\right)$.

3. 解 $F(t)=\iiint_{\Omega_t}[z^2+f(x^2+y^2)]\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z=\int_0^1\mathrm{d}z\int_0^{2\pi}\mathrm{d}\theta\int_0^t[z^2+f(r^2)]r\,\mathrm{d}r$

$$=2\pi\int_0^1\mathrm{d}z\int_0^t[z^2+f(r^2)]r\,\mathrm{d}r=2\pi\int_0^t\left[\frac{r}{3}+rf(r^2)\right]\mathrm{d}r,$$

则$\lim\limits_{t\to0^+}\dfrac{F(t)}{t^2}=2\pi\lim\limits_{t\to0^+}\dfrac{\int_0^t\left[\frac{r}{3}+rf(r^2)\right]\mathrm{d}r}{t^2}=\pi\lim\limits_{t\to0^+}\dfrac{\frac{t}{3}+tf(t^2)}{t}=\dfrac{\pi}{3}$.

4. 解 设曲面Σ、$z=0$与$z=\pi$围成的几何体为Ω，任取$z\in[0,\pi]$，截口椭圆的面积为$S(z)=\pi\sqrt{z\sin^2 z}\cdot\dfrac{\sqrt{z\sin^2 z}}{2}=\dfrac{\pi}{2}z\sin^2 z$，则所求体积为

$$V=\iiint_{\Omega}\mathrm{d}V=\int_0^{\pi}\mathrm{d}z\iint_{D_z}\mathrm{d}x\,\mathrm{d}y=\frac{\pi}{2}\int_0^{\pi}z\sin^2 z\,\mathrm{d}z=\frac{\pi}{2}\cdot\frac{\pi}{2}\cdot2\int_0^{\frac{\pi}{2}}\sin^2 z\,\mathrm{d}z=\frac{\pi^3}{8}.$$

5. 解 令$F(x)=\int_0^x f(t)\mathrm{d}t$，$F(1)=A$，$F(0)=0$，则

$$\int_0^1\mathrm{d}x\int_0^x\mathrm{d}y\int_0^y f(x)f(y)f(z)\mathrm{d}z=\int_0^1 f(x)\mathrm{d}x\int_0^x f(y)\mathrm{d}y\int_0^y f(z)\mathrm{d}z$$

$$=\int_0^1 f(x)\mathrm{d}x\int_0^x f(y)F(y)\mathrm{d}y=\int_0^1 f(x)\mathrm{d}x\int_0^x F(y)\mathrm{d}[F(y)]$$

$$=\frac{1}{2}\int_0^1 f(x)F^2(x)\mathrm{d}x=\frac{1}{6}F^3(x)\Big|_0^1=\frac{A^3}{6}.$$

6. 解 $L:\dfrac{x}{1}=\dfrac{y+1}{1}=\dfrac{z-1}{1}$，参数形式为$L:\begin{cases}x=t,\\y=-1+t,\\z=1+t,\end{cases}$

$$\mathrm{d}s=\sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2+\left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2+\left(\frac{\mathrm{d}z}{\mathrm{d}t}\right)^2}\,\mathrm{d}t=\sqrt{3}\,\mathrm{d}t,$$

则$\int_L(x+y+z)\mathrm{d}s=3\sqrt{3}\int_0^1 t\,\mathrm{d}t=\dfrac{3\sqrt{3}}{2}$.

7. 解 由对称性可知$\oint_L(x^2+2y^2+z)\mathrm{d}s=\oint_L(x^2+2y^2)\mathrm{d}s=\oint_L(x^2+y^2+z^2)\mathrm{d}s$

$$=\oint_L a^2\mathrm{d}s=a^2\oint_L\mathrm{d}s=2\pi a^3.$$

8. 解 L的极坐标形式为$L:r^2=a^2\cos 2\theta$，

$$\mathrm{d}s=\sqrt{r^2(\theta)+r'^2(\theta)}\,\mathrm{d}\theta=\frac{a}{\sqrt{\cos 2\theta}}\mathrm{d}\theta,$$

$$\int_L|y|\mathrm{d}s=4\int_0^{\frac{\pi}{4}}a\sqrt{\cos 2\theta}\sin\theta\cdot\frac{a}{\sqrt{\cos 2\theta}}\mathrm{d}\theta=4a^2\int_0^{\frac{\pi}{4}}\sin\theta\,\mathrm{d}\theta=2a^2(2-\sqrt{2}).$$

9. 解 $P(x,y)=ye^x-e^{-y}+y$，$Q(x,y)=xe^{-y}+e^x$.

$$\frac{\partial Q}{\partial x}=e^{-y}+e^x,\quad\frac{\partial P}{\partial y}=e^x+e^{-y}+1,\quad\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}=-1,$$

令$L_0:y=0$(起点$x=2$，终点$x=0$)，则

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$$\int_L (ye^x - e^{-y} + y)dx + (xe^{-y} + e^x)dy$$
$$= \left(\oint_{L+L_0} - \int_{L_0}\right)(ye^x - e^{-y} + y)dx + (xe^{-y} + e^x)dy,$$
$$\oint_{L+L_0} (ye^x - e^{-y} + y)dx + (xe^{-y} + e^x)dy$$
$$= \iint_D dx\,dy = \int_0^2 dx \int_0^{x(2-x)} dy = \int_0^2 x(2-x)dx = \frac{4}{3},$$
$$\int_{L_0} (ye^x - e^{-y} + y)dx + (xe^{-y} + e^x)dy = \int_2^0 -dx = 2,$$

于是$\int_L (ye^x - e^{-y} + y)dx + (xe^{-y} + e^x)dy = \frac{4}{3} - 2 = -\frac{2}{3}$.

10. 解 $P(x,y) = xf(x^2+y^2)$，$Q(x,y) = yf(x^2+y^2)$.

因为$\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y} = 2xyf'(x^2+y^2)$，所以曲线积分与路径无关，故

$$I = \int_L f(x^2+y^2)(x\,dx + y\,dy) = \frac{1}{2}\int_{(0,0)}^{(2,0)} f(x^2+y^2)d(x^2+y^2) = \frac{1}{2}\int_0^4 f(t)dt = 1.$$

11. 解 $\operatorname{div} \boldsymbol{A} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = 6x^2yz - 2x^2yz - 2x^2yz = 2x^2yz$.

$\frac{\partial}{\partial x}(\operatorname{div}\boldsymbol{A})\Big|_M = 8$，$\frac{\partial}{\partial y}(\operatorname{div}\boldsymbol{A})\Big|_M = 4$，$\frac{\partial}{\partial z}(\operatorname{div}\boldsymbol{A})\Big|_M = 2$，$\cos\alpha = \frac{2}{3}$，$\cos\beta = \frac{2}{3}$，$\cos\gamma = -\frac{1}{3}$.

则$\frac{\partial}{\partial \boldsymbol{l}}(\operatorname{div}\boldsymbol{A})\Big|_M = 8\times\frac{2}{3} + 4\times\frac{2}{3} + 2\times\left(-\frac{1}{3}\right) = \frac{22}{3}$.

三、解答题

1. 解 由对称性与奇偶性，得 $I = \iiint_\Omega (x + x^2\sin y + z)dV = \iiint_\Omega z\,dV$.

由切片法，得 $I = \iiint_\Omega z\,dV = \int_0^2 z\,dz \iint_{x^2+y^2\leqslant 2z-z^2} dx\,dy = \pi\int_0^2 z(2z-z^2)dz = \frac{4\pi}{3}$.

2. 解 由对称性与奇偶性，得

$$I = \iiint_\Omega (x^2z + x\sin^2 y)dV = \iiint_\Omega x^2z\,dV = \iiint_\Omega y^2z\,dV$$
$$= \frac{1}{2}\iiint_\Omega (x^2+y^2)z\,dV.$$

方法一 $\Omega = \{(x,y,z) \mid (x,y)\in D_{xy}, \sqrt{x^2+y^2} \leqslant z \leqslant \sqrt{2-x^2-y^2}\}$.

其中 $D_{xy} = \{(x,y) \mid x^2+y^2 \leqslant 1\}$，则

$$I = \frac{1}{2}\iiint_\Omega (x^2+y^2)z\,dV = \frac{1}{2}\iint_{D_{xy}} (x^2+y^2)dx\,dy \int_{\sqrt{x^2+y^2}}^{\sqrt{2-x^2-y^2}} z\,dz$$
$$= \frac{1}{2}\iint_{D_{xy}} (x^2+y^2)(1-x^2-y^2)dx\,dy = \frac{1}{2}\int_0^{2\pi} d\theta \int_0^1 r^3(1-r^2)dr = \frac{\pi}{12}.$$

方法二 令$\begin{cases} x = r\cos\theta\sin\varphi, \\ y = r\sin\theta\sin\varphi, \\ z = r\cos\varphi \end{cases}$，$\left(0\leqslant\theta\leqslant 2\pi, 0\leqslant\varphi\leqslant\frac{\pi}{4}, 0\leqslant r\leqslant\sqrt{2}\right)$，则

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$$I=\frac{1}{2}\int_0^{2\pi}\mathrm{d}\theta\int_0^{\frac{\pi}{4}}\mathrm{d}\varphi\int_0^{\sqrt{2}}r^2\sin\varphi\cdot r^2\sin^2\varphi\cdot r\cos\varphi\,\mathrm{d}r=\pi\int_0^{\frac{\pi}{4}}\sin^3\varphi\,\mathrm{d}(\sin\varphi)\int_0^{\sqrt{2}}r^5\,\mathrm{d}r=\frac{\pi}{12}.$$

3. 证明 令 $f(x,y,z)=x+2y-2z+5$，因为

$$f'_x=1\neq 0,f'_y=2\neq 0,f'_z=-2\neq 0,$$

所以 $f(x,y,z)$ 在区域 Ω 的边界 $x^2+y^2+z^2=1$ 上取到最大值和最小值.

令 $F(x,y,z,\lambda)=x+2y-2z+5+\lambda(x^2+y^2+z^2-1)$.

由 $\begin{cases}F'_x=1+2\lambda x=0,\\F'_y=2+2\lambda y=0,\\F'_z=-2+2\lambda z=0,\\F'_\lambda=x^2+y^2+z^2-1=0,\end{cases}$ 得驻点为 $P_1\left(\frac{1}{3},\frac{2}{3},-\frac{2}{3}\right),P_2\left(-\frac{1}{3},-\frac{2}{3},\frac{2}{3}\right)$.

因为 $f(P_1)=8,f(P_2)=2$，所以 $\sqrt[3]{f(x,y,z)}$ 在 Ω 上的最大值与最小值分别为 2 和 $\sqrt[3]{2}$，于是

$$2\iiint_\Omega\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z=\frac{8\pi}{3},\sqrt[3]{2}\iiint_\Omega\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z=\frac{4\sqrt[3]{2}\pi}{3},$$

$$\frac{4\sqrt[3]{2}\pi}{3}\leqslant\iiint_\Omega\sqrt[3]{x+2y-2z+5}\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\leqslant\frac{8\pi}{3}.$$

4. 解 令 $P=\frac{x-y}{x^2+4y^2},Q=\frac{x+4y}{x^2+4y^2},\frac{\partial Q}{\partial x}=\frac{\partial P}{\partial y}$，令 $C:x^2+4y^2=r^2(r>0)$ 逆时针且 C 在曲线 L 内，则

$$\begin{aligned}\oint_L\frac{(x-y)\mathrm{d}x+(x+4y)\mathrm{d}y}{x^2+4y^2}&=\left(\oint_{L+C^-}+\oint_C\right)\frac{(x-y)\mathrm{d}x+(x+4y)\mathrm{d}y}{x^2+4y^2}\\&=\frac{1}{r^2}\oint_C(x-y)\mathrm{d}x+(x+4y)\mathrm{d}y\\&=\frac{1}{r^2}\iint_D2\mathrm{d}\sigma=2\cdot\frac{1}{r^2}\cdot\pi\cdot r\cdot\frac{r}{2}=\pi.\end{aligned}$$

5. 解 **方法一** 设由 L 所围成的平面为 Σ，按右手准则，Σ 取上侧.

$$\boldsymbol{n}=\{0,-1,1\},\cos\alpha=0,\cos\beta=-\frac{1}{\sqrt{2}},\cos\gamma=\frac{1}{\sqrt{2}}.$$

由斯托克斯公式，得 $\oint_C xyz\,\mathrm{d}z=\frac{1}{\sqrt{2}}\iint_\Sigma\begin{vmatrix}0&-1&1\\\frac{\partial}{\partial x}&\frac{\partial}{\partial y}&\frac{\partial}{\partial z}\\0&0&xyz\end{vmatrix}\mathrm{d}S=\frac{1}{\sqrt{2}}\iint_\Sigma yz\,\mathrm{d}S.$

$\Sigma:z=y((x,y)\in D_{xy})$，其中 $D_{xy}:x^2+2y^2\leqslant 1$. $\mathrm{d}S=\sqrt{1+z'^2_x+z'^2_y}\,\mathrm{d}x\,\mathrm{d}y=\sqrt{2}\,\mathrm{d}x\,\mathrm{d}y$.

故 $\oint_C xyz\,\mathrm{d}z=\iint_{D_{xy}}y^2\,\mathrm{d}x\,\mathrm{d}y=\frac{\sqrt{2}}{16}\pi$.

方法二 令 $C:\begin{cases}x=\cos t,\\y=\frac{\sqrt{2}}{2}\sin t,\\z=\frac{\sqrt{2}}{2}\sin t,\end{cases}$ 则

$$\oint_C xyz\,\mathrm{d}z=\frac{\sqrt{2}}{4}\int_0^{2\pi}\cos^2t\sin^2t\,\mathrm{d}t=\sqrt{2}\int_0^{\frac{\pi}{2}}(1-\sin^2t)\sin^2t\,\mathrm{d}t=\frac{\sqrt{2}}{16}\pi.$$

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6. 解 设由 L 所围成的平面为 Σ，按右手准则，Σ 取上侧.

$\boldsymbol{n}=\{0,-3,1\}$，$\cos\alpha=0$，$\cos\beta=-\dfrac{3}{\sqrt{10}}$，$\cos\gamma=\dfrac{1}{\sqrt{10}}$，由斯托克斯公式，得

$$\oint_L yz\,dx+3xz\,dy-xy\,dz=\frac{1}{\sqrt{10}}\iint_\Sigma\begin{vmatrix}0&-3&1\\ \frac{\partial}{\partial x}&\frac{\partial}{\partial y}&\frac{\partial}{\partial z}\\ yz&3xz&-xy\end{vmatrix}dS=\frac{1}{\sqrt{10}}\iint_\Sigma\begin{vmatrix}0&-3&1\\ \frac{\partial}{\partial x}&\frac{\partial}{\partial y}&\frac{\partial}{\partial z}\\ yz&3xz&-xy\end{vmatrix}dS$$

$$=\frac{2}{\sqrt{10}}\iint_\Sigma(-3y+z)\,dS=\frac{2}{\sqrt{10}}\iint_\Sigma dS.$$

因为 $dS=\sqrt{1+z_x'^2+z_y'^2}\,dx\,dy=\sqrt{10}\,dx\,dy$，$D_{xy}:x^2+y^2\leqslant 4y$，所以

$$\oint_L yz\,dx+3xz\,dy-xy\,dz=2\iint_{D_{xy}}dx\,dy=8\pi.$$

7. 解 取平面 $x+y+z=\dfrac{3}{2}$ 上被折线 C 所围的上侧部分为 Σ，其法向量的方向余弦为

$$\cos\alpha=\cos\beta=\cos\gamma=\frac{\sqrt{3}}{3}.$$

设 D_{xy} 表示曲面 Σ 在平面 xOy 上的投影区域，其面积为 $A=\dfrac{3}{4}$，由斯托克斯公式，得

$$\left|\oint_C(z^2-y^2)dx+(x^2-z^2)dy+(y^2-x^2)dz\right|$$

$$=\iint_\Sigma\begin{vmatrix}\cos\alpha&\cos\beta&\cos\gamma\\ \frac{\partial}{\partial x}&\frac{\partial}{\partial y}&\frac{\partial}{\partial z}\\ z^2-y^2&x^2-z^2&y^2-x^2\end{vmatrix}dS=\frac{4}{\sqrt{3}}\iint_\Sigma(x+y+z)\,dS$$

$$=\frac{6}{\sqrt{3}}\iint_\Sigma dS=\frac{6}{\sqrt{3}}\iint_{D_{xy}}\sqrt{1+z_x'^2+z_y'^2}\,dx\,dy=\frac{6}{\sqrt{3}}\cdot\sqrt{3}\cdot\frac{3}{4}=\frac{9}{2}.$$

8. 解 令 $P(x,y)=\dfrac{x+y}{x^2+y^2}$，$Q(x,y)=-\dfrac{x-y}{x^2+y^2}=\dfrac{y-x}{x^2+y^2}$，显然 $\dfrac{\partial Q}{\partial x}=\dfrac{\partial P}{\partial y}$.

令 $L_r:x^2+y^2=r^2$，其中 $r>0$，L_r 在 L 内，方向取逆时针，由格林公式得

$$\oint_{L+L_r^-}\frac{(x+y)dx-(x-y)dy}{x^2+y^2}=\iint_D\left(\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}\right)dx\,dy=0,$$

其中 D 为 L 与 L_r^- 所围成的平面区域，则

$$I=\oint_L\frac{(x+y)dx-(x-y)dy}{x^2+y^2}=\oint_{L_r}\frac{(x+y)dx-(x-y)dy}{x^2+y^2},$$

$$\oint_{L_r}\frac{(x+y)dx-(x-y)dy}{x^2+y^2}$$

$$=\int_0^{2\pi}\frac{-r(r\cos\theta+r\sin\theta)\sin\theta-r(r\cos\theta-r\sin\theta)\cos\theta}{r^2}d\theta=-2\pi,$$

所以 $I=\oint_L\dfrac{(x+y)dx-(x-y)dy}{x^2+y^2}=-2\pi.$

9. 解 $I=\oint_L\dfrac{y\,dx}{(2-x)^2+y^2}+\dfrac{(2-x)dy}{(2-x)^2+y^2}+\oint_L\dfrac{y\,dx}{(2+x)^2+y^2}-\dfrac{(2+x)dy}{(2+x)^2+y^2}$

$=I_1+I_2$，

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显然曲线积分 I_1, I_2 都满足柯西-黎曼条件.

① 当(2,0),(−2,0) 都在 L 所围成的区域之外时,$I_1 = I_2 = 0$,因此 $I = 0$;

② 当(2,0),(−2,0) 都在 L 所围成的区域之内时,分别以这两个点为中心以 r_1, r_2 为半径作圆 C_1, C_2,使它们都在 L 内,则 $I_1 = \oint_{C_1} \frac{y\mathrm{d}x}{(2-x)^2+y^2} + \frac{(2-x)\mathrm{d}y}{(2-x)^2+y^2} = -2\pi$,同理,$I_2 = -2\pi$,因此 $I = -4\pi$;

③ 当(2,0),(−2,0) 有一个点在 L 围成的区域内,一个点在 L 围成的区域外时,$I = -2\pi$.

10. 证明 $P\mathrm{d}x + Q\mathrm{d}y = \{P, Q\} \cdot \{\mathrm{d}x, \mathrm{d}y\}$.

因为 $|\boldsymbol{a} \cdot \boldsymbol{b}| \leqslant |\boldsymbol{a}||\boldsymbol{b}|$,所以

$$|P\mathrm{d}x + Q\mathrm{d}y| \leqslant \sqrt{P^2+Q^2} \cdot \sqrt{(\mathrm{d}x)^2+(\mathrm{d}y)^2} \leqslant M\mathrm{d}s,$$

于是 $\left|\int_L P\mathrm{d}x + Q\mathrm{d}y\right| \leqslant \int_L |P\mathrm{d}x + Q\mathrm{d}y| \leqslant \int_L M\mathrm{d}s = M\int_L \mathrm{d}s = Ml.$

11. 解 因为曲线积分与路径无关,所以 $\cos y = f'_y(x, y)$,则 $f(x, y) = \sin y + C(x)$,而

$$\int_{(0,0)}^{(t,t^2)} f(x, y)\mathrm{d}x + x\cos y\mathrm{d}y = t^2,$$

即 $\int_0^t C(x)\mathrm{d}x + \int_0^{t^2} t\cos y\mathrm{d}y = t^2$,两边对 t 求导数,得

$$C(t) = 2t - \sin t^2 - 2t^2\cos t^2,$$

于是 $f(x, y) = \sin y + 2x - \sin x^2 - 2x^2\cos x^2$.

12. 解 因为曲线积分与路径无关,所以

$$f''(x) = 3f'(x) - 2f(x) + x\mathrm{e}^{2x} \Rightarrow f''(x) - 3f'(x) + 2f(x) = x\mathrm{e}^{2x},$$

由特征方程 $\lambda^2 - 3\lambda + 2 = 0$,得 $\lambda_1 = 1, \lambda_2 = 2$.

则方程 $f''(x) - 3f'(x) + 2f(x) = 0$ 的通解为 $f(x) = C_1\mathrm{e}^x + C_2\mathrm{e}^{2x}$.

令特解 $f_0(x) = x(ax+b)\mathrm{e}^{2x}$,代入原微分方程得 $a = \frac{1}{2}, b = -1$.

故所求 $f(x) = C_1\mathrm{e}^x + C_2\mathrm{e}^{2x} + \left(\frac{x^2}{2} - x\right)\mathrm{e}^{2x}$.

13. 解 曲面 $2z = x^2 + y^2$ 上任一点 (x, y, z) 指向上侧的法向量为 $\boldsymbol{n} = \{-x, -y, 1\}$,法向量的方向余弦为

$$\cos\alpha = -\frac{x}{\sqrt{1+x^2+y^2}}, \quad \cos\beta = -\frac{y}{\sqrt{1+x^2+y^2}}, \quad \cos\gamma = \frac{1}{\sqrt{1+x^2+y^2}},$$

则 $\iint_\Sigma [yf(x, y, z) + x]\mathrm{d}y\mathrm{d}z + [xf(x, y, z) + y]\mathrm{d}z\mathrm{d}x + [2xyf(x, y, z) + z]\mathrm{d}x\mathrm{d}y$

$$= \iint_\Sigma \{[yf(x, y, z) + x]\cos\alpha + [xf(x, y, z) + y]\cos\beta + [2xyf(x, y, z) + z]\cos\gamma\}\mathrm{d}S$$

$$= -\iint_\Sigma \frac{z}{\sqrt{1+x^2+y^2}}\mathrm{d}S = -\frac{1}{2}\iint_\Sigma \frac{x^2+y^2}{\sqrt{1+x^2+y^2}}\mathrm{d}S.$$

因为 $\mathrm{d}S = \sqrt{1 + z'^2_x + z'^2_y}\,\mathrm{d}x\mathrm{d}y = \sqrt{1+x^2+y^2}\,\mathrm{d}x\mathrm{d}y$,所以

$$\text{原式} = -\frac{1}{2}\iint_D (x^2+y^2)\mathrm{d}x\mathrm{d}y = -\frac{1}{2}\int_0^{2\pi}\mathrm{d}\theta\int_2^4 r^3\mathrm{d}r = -60\pi.$$

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14. 解 设 $\Sigma_1: z=\sqrt{1-x^2-y^2}$，取上侧，由对称性得 $\oiint\limits_{\Sigma}\frac{\mathrm{d}z\,\mathrm{d}x}{\cos^2 y}=0$.

由轮换对称性得 $\oiint\limits_{\Sigma}\frac{\mathrm{d}y\,\mathrm{d}z}{x\cos^2 x}=\oiint\limits_{\Sigma}\frac{\mathrm{d}x\,\mathrm{d}y}{z\cos^2 z}$，则

$$I=\oiint\limits_{\Sigma}\frac{2\mathrm{d}y\,\mathrm{d}z}{x\cos^2 x}+\frac{\mathrm{d}z\,\mathrm{d}x}{\cos^2 y}-\frac{\mathrm{d}x\,\mathrm{d}y}{z\cos^2 z}=\oiint\limits_{\Sigma}\frac{\mathrm{d}x\,\mathrm{d}y}{z\cos^2 z}=2\iint\limits_{\Sigma_1}\frac{\mathrm{d}x\,\mathrm{d}y}{z\cos^2 z}$$

$$=2\iint\limits_{x^2+y^2\leqslant 1}\frac{\mathrm{d}x\,\mathrm{d}y}{\sqrt{1-x^2-y^2}\cos^2\sqrt{1-x^2-y^2}}$$

$$=4\pi\int_0^1\frac{r\,\mathrm{d}r}{\sqrt{1-r^2}\cos^2\sqrt{1-r^2}}=-4\pi\int_0^1\sec^2\sqrt{1-r^2}\,\mathrm{d}(\sqrt{1-r^2})$$

$$=-4\pi\tan\sqrt{1-r^2}\Big|_0^1=4\pi\tan 1.$$

15. 解 $\iint\limits_{\Sigma}\frac{\mathrm{d}S}{x^2+y^2+z^2}=\iint\limits_{\Sigma}\frac{\mathrm{d}S}{a^2+z^2}$.

令 $\Sigma_1: y=-\sqrt{a^2-x^2}$，$\Sigma_2: y=\sqrt{a^2-x^2}$，$D_{xz}=\{(x,z)\mid -a\leqslant x\leqslant a,-a\leqslant z\leqslant a\}$，

$$\mathrm{d}S=\sqrt{1+\left(\frac{\partial y}{\partial x}\right)^2+\left(\frac{\partial y}{\partial z}\right)^2}\,\mathrm{d}x\,\mathrm{d}z=\frac{a}{\sqrt{a^2-x^2}}\mathrm{d}x\,\mathrm{d}z,$$

由对称性得

$$\iint\limits_{\Sigma}\frac{\mathrm{d}S}{x^2+y^2+z^2}=\iint\limits_{\Sigma}\frac{\mathrm{d}S}{a^2+z^2}=2\iint\limits_{\Sigma_1}\frac{\mathrm{d}S}{a^2+z^2}=2a\iint\limits_{D_{xz}}\frac{\mathrm{d}x\,\mathrm{d}z}{(a^2+z^2)\sqrt{a^2-x^2}}$$

$$=2a\int_{-a}^{a}\frac{\mathrm{d}x}{\sqrt{a^2-x^2}}\int_{-a}^{a}\frac{\mathrm{d}z}{a^2+z^2}=8a\int_0^a\frac{\mathrm{d}x}{\sqrt{a^2-x^2}}\int_0^a\frac{\mathrm{d}z}{a^2+z^2}$$

$$=8a\cdot\frac{\pi}{2}\cdot\frac{1}{a}\cdot\frac{\pi}{4}=\pi^2.$$

16. 解 由对称性得 $\iint\limits_{\Sigma}x^2\mathrm{d}S=\iint\limits_{\Sigma}y^2\mathrm{d}S=\iint\limits_{\Sigma}z^2\mathrm{d}S=\frac{1}{3}\iint\limits_{\Sigma}(x^2+y^2+z^2)\mathrm{d}S$，而

$$\iint\limits_{\Sigma}(x^2+y^2+z^2)\mathrm{d}S=a^2\iint\limits_{\Sigma}\mathrm{d}S=4\pi a^4,$$

所以 $\iint\limits_{\Sigma}(x^2+4y^2+9z^2)\mathrm{d}S=14\iint\limits_{\Sigma}x^2\mathrm{d}S=14\cdot\frac{1}{3}\cdot 4\pi a^4=\frac{56\pi a^4}{3}$.

17. 解 曲面 $\Sigma: z=1-x^2-y^2(z\geqslant 0)$，补充曲面 $\Sigma_0: z=0(x^2+y^2\leqslant 1)$，取下侧，由高斯公式得

$$\oiint\limits_{\Sigma+\Sigma_0}(x^3+z)\mathrm{d}y\,\mathrm{d}z+(y^3+x)\mathrm{d}z\,\mathrm{d}x+\mathrm{d}x\,\mathrm{d}y=3\iiint\limits_{\Omega}(x^2+y^2)\mathrm{d}V$$

$$=3\iint\limits_{x^2+y^2\leqslant 1}(x^2+y^2)\mathrm{d}x\,\mathrm{d}y\int_0^{1-x^2-y^2}\mathrm{d}z=3\iint\limits_{x^2+y^2\leqslant 1}(x^2+y^2)(1-x^2-y^2)\mathrm{d}x\,\mathrm{d}y$$

$$=3\int_0^{2\pi}\mathrm{d}\theta\int_0^1 r^3(1-r^2)\mathrm{d}r=\frac{\pi}{2},$$

$$\iint\limits_{\Sigma_0}(x^3+z)\mathrm{d}y\,\mathrm{d}z+(y^3+x)\mathrm{d}z\,\mathrm{d}x+\mathrm{d}x\,\mathrm{d}y=\iint\limits_{\Sigma_0}\mathrm{d}x\,\mathrm{d}y=-\iint\limits_{x^2+y^2\leqslant 1}\mathrm{d}x\,\mathrm{d}y=-\pi,$$

则 $\iint\limits_{\Sigma}(x^3+z)\mathrm{d}y\,\mathrm{d}z+(y^3+x)\mathrm{d}z\,\mathrm{d}x+\mathrm{d}x\,\mathrm{d}y=\frac{3\pi}{2}$.

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18. 证明 令 $\begin{cases}x=r\cos\theta,\\ y=r\sin\theta.\end{cases}$

由 $\dfrac{\partial f}{\partial r}=\dfrac{\partial f}{\partial x}\cdot\cos\theta+\dfrac{\partial f}{\partial y}\cdot\sin\theta$，得 $r\dfrac{\partial f}{\partial r}=\dfrac{\partial f}{\partial x}\cdot r\cos\theta+\dfrac{\partial f}{\partial y}\cdot r\sin\theta=x\dfrac{\partial f}{\partial x}+y\dfrac{\partial f}{\partial y}$.

于是 $I=\iint\limits_{D_r}\dfrac{x\dfrac{\partial f}{\partial x}+y\dfrac{\partial f}{\partial y}}{x^2+y^2}\mathrm{d}x\,\mathrm{d}y=\int_0^{2\pi}\mathrm{d}\theta\int_r^1\dfrac{r\dfrac{\partial f}{\partial r}}{r^2}\cdot r\,\mathrm{d}r$

$=\int_0^{2\pi}[f(\cos\theta,\sin\theta)-f(r\cos\theta,r\sin\theta)]\mathrm{d}\theta=-\int_0^{2\pi}f(r\cos\theta,r\sin\theta)\mathrm{d}\theta$.

再根据积分中值定理得 $I=-2\pi f(r\cos\xi,r\sin\xi)$，其中 ξ 是介于 0 与 2π 之间的值.

故原式 $=-\dfrac{1}{2\pi}\cdot\lim\limits_{r\to0}[-2\pi f(r\cos\xi,r\sin\xi)]=\lim\limits_{r\to0}f(r\cos\xi,r\sin\xi)=f(0,0)$.

19. 解 $f\cdot g=v\left(\dfrac{\partial u}{\partial x}-\dfrac{\partial u}{\partial y}\right)+u\left(\dfrac{\partial v}{\partial x}-\dfrac{\partial v}{\partial y}\right)=v\dfrac{\partial u}{\partial x}+u\dfrac{\partial v}{\partial x}-\left(v\dfrac{\partial u}{\partial y}+u\dfrac{\partial v}{\partial y}\right)$

$=\dfrac{\partial(uv)}{\partial x}-\dfrac{\partial(uv)}{\partial y}$,

则 $\iint\limits_D f\cdot g\,\mathrm{d}\sigma=\iint\limits_D\left[\dfrac{\partial(uv)}{\partial x}-\dfrac{\partial(uv)}{\partial y}\right]\mathrm{d}\sigma=\oint_L uv\,\mathrm{d}x+uv\,\mathrm{d}y=\oint_L y\,\mathrm{d}x+y\,\mathrm{d}y$

$=\int_0^{2\pi}(-\sin^2\theta+\sin\theta\cos\theta)\mathrm{d}\theta=-\pi$（其中 L 为单位圆周的正向）.

20. 解 任取 $M(x,y)\in L$，$r=\sqrt{x^2+(y-1)^2}$.

两质点的引力大小为 $|\boldsymbol{F}|=\dfrac{k}{r^2}=\dfrac{k}{x^2+(y-1)^2}$,

$$\overrightarrow{MA}=\{-x,1-y\},\quad \boldsymbol{F}^0=\overrightarrow{MA}^0=\frac{1}{\sqrt{x^2+(y-1)^2}}\{-x,1-y\},$$

则 $\boldsymbol{F}=|\boldsymbol{F}|\boldsymbol{F}^0=\dfrac{k}{[x^2+(y-1)^2]^{\frac{3}{2}}}\{-x,1-y\}$，则 $W=\int_L P(x,y)\mathrm{d}x+Q(x,y)\mathrm{d}y$，其中

$$P(x,y)=-\frac{kx}{[x^2+(y-1)^2]^{\frac{3}{2}}},\quad Q(x,y)=\frac{k(1-y)}{[x^2+(y-1)^2]^{\frac{3}{2}}}.$$

因为 $\dfrac{\partial Q}{\partial x}=\dfrac{\partial P}{\partial y}=\dfrac{3kx(y-1)}{[x^2+(y-1)^2]^{\frac{5}{2}}}$，所以曲线积分与路径无关，从而

$$W=\int_{\overline{BO}}P(x,y)\mathrm{d}x+Q(x,y)\mathrm{d}y=-k\int_2^0\frac{x}{(x^2+1)^{\frac{3}{2}}}\mathrm{d}x$$

$$=-\frac{k}{\sqrt{1+x^2}}\Big|_0^2=k\left(1-\frac{\sqrt{5}}{5}\right).$$

21. 解 设原点 O 到点 $M(\xi,\eta,\zeta)$ 的直线为 L，L 的参数方程为 $L:\begin{cases}x=\xi t,\\ y=\eta t,\\ z=\zeta t\end{cases}$（起点 $t=0$，终点 $t=1$），则

$$W=\int_L yz\,\mathrm{d}x+xz\,\mathrm{d}y+xy\,\mathrm{d}z=\int_0^1 3\xi\eta\zeta t^2\,\mathrm{d}t=\xi\eta\zeta.$$

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令 $F(\xi,\eta,\zeta)=\xi\eta\zeta+\lambda\left(\frac{\xi^2}{a^2}+\frac{\eta^2}{b^2}+\frac{\zeta^2}{c^2}-1\right)$，由$\begin{cases}F'_\xi=\eta\zeta+\frac{2\lambda\xi}{a^2}=0,\\F'_\eta=\xi\zeta+\frac{2\lambda\eta}{b^2}=0,\\F'_\zeta=\xi\eta+\frac{2\lambda\zeta}{c^2}=0,\\F'_\lambda=\frac{\xi^2}{a^2}+\frac{\eta^2}{b^2}+\frac{\zeta^2}{c^2}-1=0,\end{cases}$ 得

$$\xi=\frac{a}{\sqrt{3}},\eta=\frac{b}{\sqrt{3}},\zeta=\frac{c}{\sqrt{3}}.$$

所以当点 M 的坐标为$\left(\frac{a}{\sqrt{3}},\frac{b}{\sqrt{3}},\frac{c}{\sqrt{3}}\right)$时，力$\boldsymbol{F}$ 所做的功最大，且最大功为$\frac{\sqrt{3}}{9}abc$.

22. 解 设 Σ 围成的几何体为 Ω，由高斯公式得

$$I=3\iiint_\Omega(x^2+2y^2+3z^2-1)\mathrm{d}V,$$

当 Σ 为椭球面 $x^2+2y^2+3z^2=1$ 的外侧时，I 取最小，

$$\begin{aligned}I_{\min}&=\frac{3}{\sqrt{6}}\iiint_\Omega[x^2+(\sqrt{2}y)^2+(\sqrt{3}z)^2-1]\mathrm{d}x\,\mathrm{d}(\sqrt{2}y)\,\mathrm{d}(\sqrt{3}z)\\&=\frac{3}{\sqrt{6}}\iiint_{x^2+y^2+z^2\leqslant1}(x^2+y^2+z^2-1)\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\\&=\frac{3}{\sqrt{6}}\left(\int_0^{2\pi}\mathrm{d}\theta\int_0^{\pi}\sin\varphi\,\mathrm{d}\varphi\int_0^1r^4\mathrm{d}r-\frac{4}{3}\pi\right)=-\frac{8\pi}{5\sqrt{6}}.\end{aligned}$$

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